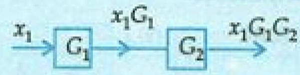
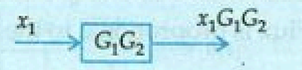
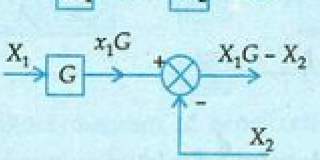
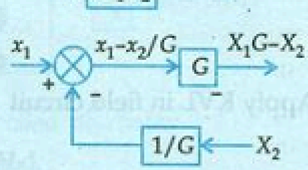
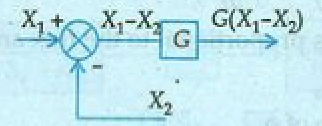
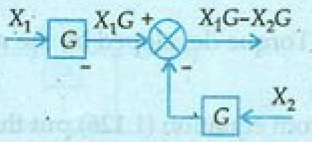
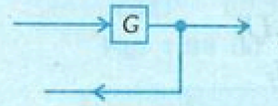
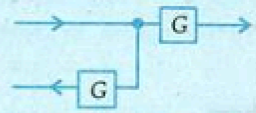
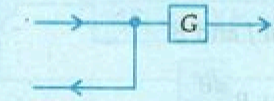
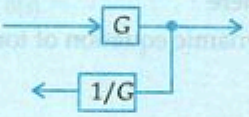
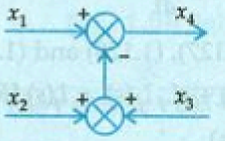
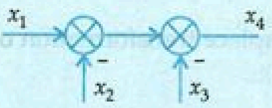
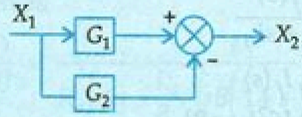
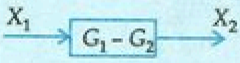
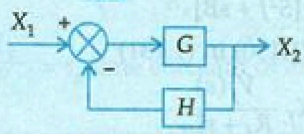
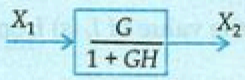
Rule	Original Diagram	Equivalent Diagram
1. Blocks in cascade		
2. Moving a summing point ahead of a block		
3. Moving a summing point beyond a block		
4. Moving a take-off point ahead of a block		
5. Moving a take-off point beyond a block		
6. Rearrangement of summing point		
7. Eliminating a forward loop		
8. Eliminating feedback loop		

1.24. SIGNAL FLOW GRAPH

The process of block diagram reduction technique is time consuming because at every stage modified block diagram is to be redrawn. A simple method was developed by S.J. Mason which is known as signal flow graph. This method is very simple and does not require any reduction technique. Signal flow graph is applicable to the linear systems.

A signal flow graph is a diagram which represents a set of simultaneous equations.

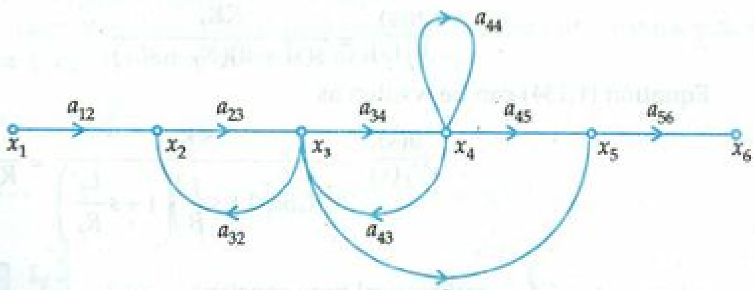


Fig. 1.101

Signal flow graph consists of nodes and these nodes are connected by a directed line called branches. Every branch of signal flow graph having an arrow, which reprsents the flow of signal.

The following terms are associated with the signal flow graph.

1. **Input node or source node :** An input node is a node which has only outgoing branches. For example x_1 is the input node.
2. **Output node or sink node :** An output node is a node that has only one or more incoming branches.*e.g.* x_6 is the output node.
3. **Mixed nodes :** A node having incoming and outgoing branches is known as mixed nodes. For example x_2, x_3, x_4, and x_5 are the mixed nodes.
4. **Transmittance :** Transmittance also known as transfer function, which is normally written on the branch near the arrow. For example a_{12}, a_{23} etc.
5. **Forward path :** Forward path is a path which originates from the input node and terminates at the output node and along which no node is traversed more than once.

   For example in Fig 1.101 there are two forward paths.

   1. x_1 to x_2 to x_3 to x_4 to x_5 to x_6
   2. x_1 to x_2 to x_3 to x_5 to x_6
6. **Loop :** Loop is a path that originates and terminates on the same node and along which no other node is traversed more than once.

   For example $\quad x_2$ to x_3 to x_2

   x_3 to x_4 to x_3
7. **Self loop :** It is a path which originates and terminates on the same node. For example x_4 to x_4
8. **Path gain :** The product of the branch gains along the path is called path gain. For example the gain of the path x_1 to x_2 to x_3 to x_4 to x_5 to x_6 is $a_{12}\,a_{23}\,a_{34}\,a_{45}\,a_{56}$
9. **Loop gain :** The gain of the loop is known as loop gain. For example the gain of the loop x_2 to x_3 to x_2 is $a_{23}\,a_{32}$.
10. **Non-touching loops :** Non touching loops having no common nodes branch and paths. For example the loops x_2 to x_3 to x_2, and x_4 to x_4 are non-touching loops.

1.25. PROPERTIES OF SIGNAL FLOW GRAPH

1. Signal flow graph is applicable to linear time-invariant systems.
2. The signal flow is only along the direction of arrows.
3. The value of variable at each node is equal to the algebraic sum of all signals entering at that node.
4. The gain of signal flow graph is given by Mason's formula.
5. The signal gets multiplied by the branch gain when it travels along it.
6. The signal flow graph is not be the unique property of the system.

S.No.	Block Diagram	SFG
1.	Applicable to linear time invariant systems only.	Applicable to linear time invariant system.
2.	Each element is represented by block.	Each variable is represented by node.
3.	Summing point and take off points are separate.	Summing and take off points are absent.
4.	Self loop do not exist.	Self loop can be exist.
5.	It is time consuming method.	Require less time by using Mason gain formula.
6.	Block diagram is required at each & every step.	At each step it is not necessary to draw SFG.
7.	Transfer function of the element is shown inside the corresponding block.	Transfer function is shown along the branches connecting the nodes.
8.	Feedback path is present.	Feedback loops are used.

1.27. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM EQUATIONS

Consider the following sets of equations

$$y_2 = t_{21}y_1 + t_{23}y_3$$
$$y_3 = t_{32}y_2 + t_{33}y_3 + t_{31}y_1$$
$$y_4 = t_{43}y_3 + t_{42}y_2$$
$$y_5 = t_{54}y_4$$
$$y_6 = t_{65}y_5 + t_{64}\ y_4$$

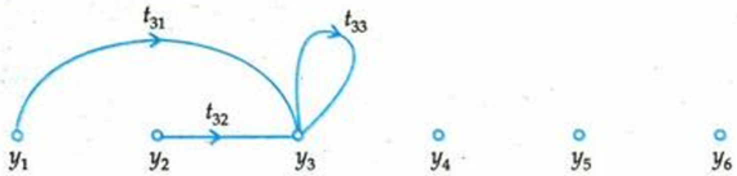
where y_1 is the input and y_6 is the output.

First of all draw the nodes. In the given example there are six nodes. From the first equation it is clear that the y_2 is the sum of two signals. Similarly, y_3 is the sum of three signals and so on. Insert the branches with proper transmittance to connect the nodes.

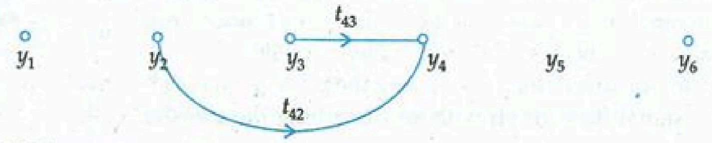
Step 1 : Draw the nodes

Step 2 : Draw the SFG for equation (1)

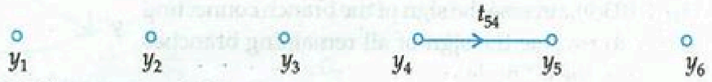
Step 3 : Draw the SFG for equation (2)



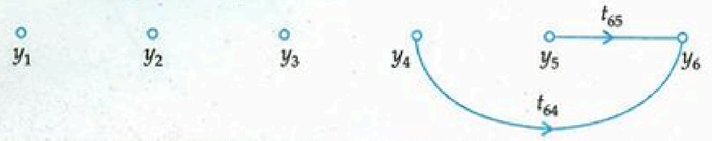
Step 4 : Draw SFG for equation (3)



Step 5 : Draw SFG for equation (4)



Step 6 : Draw SFG for equation (5)



Step 7 : Draw the complete signal flow graph with the help of above graphs.

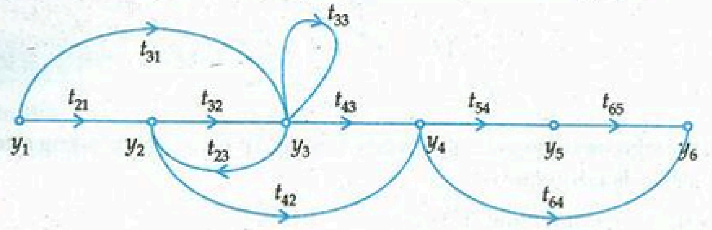


Fig. 1.102.

1.28. SIGNAL FLOW GRAPH FOR DIFFERENTIAL EQUATIONS

Consider the following defferential equation

$$y''' + 3y'' + 5y' + 2y = x \qquad \text{...(1.135)}$$

Step 1 : Solve the eqn 1.135 for the highest order

$$y''' = x - 3y'' - 5y' - 2y$$

Step 2 : Consider the left hand term (highest order derivative) as dependent variable and all other terms on right hand side as independent variables.

construct the branches of signal flow graph as shown in Fig. (1.103(*a*).

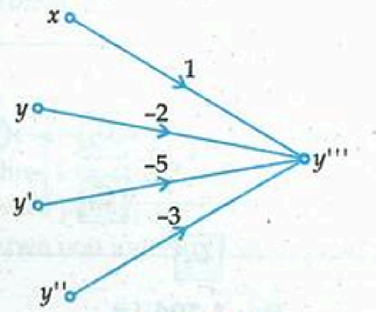


Fig. 1.103 (*a*)

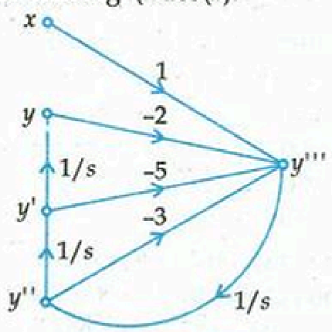
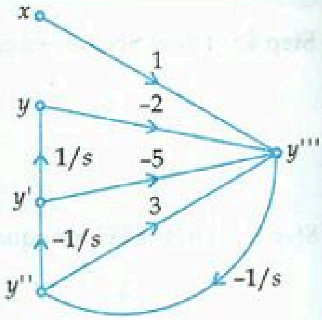


Fig. 1.103 (*b*)

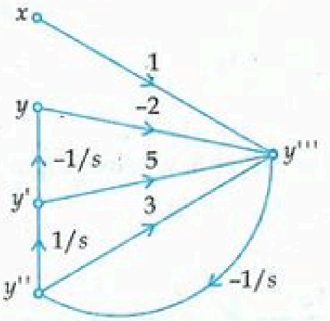
Step 3 : Connect the nodes of highest order derivative to the node whose order is lower than this and so on. The flow of the signal will be from higher node to the lower order node and transmittance will be $1/s$ as shown in Fig. 1.103 (*b*).



Step 4 : Reverse the sign of a branch connecting the p^{th} node to the q^{th} node of a signal flow graph without disturbing the transfer function.

Consider the Fig. 1.103(*b*), reverse the sign of the branch connecting y''' to y'', it is necessary to reverse the sign of all remaining branches entering as well as leaving the q^{th} node.

Similarly, reverse the sign of branch connecting y'' to y'.



By reversing the sign, we have already reverse the sign of branch connecting y' to y and therefore further reversal of sign is not required.

Step 5 : Redraw the signal flow graph (SFG).

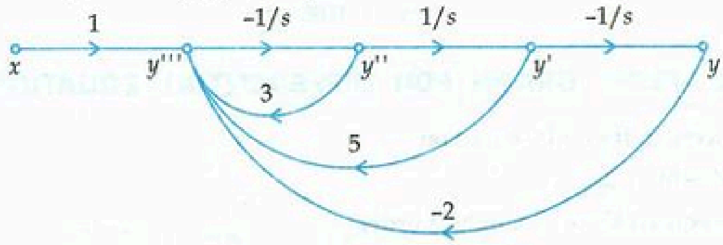


Fig. 1.103 (*c*)

1.29. CONSTRUCTION OF SIGNAL FLOW GRAPH FROM BLOCK DIAGRAM

Rules 1. All variables, summing points and take off points are represented by nodes.

2. If a summing point is placed before a take off point in the dirction of signal flow, in such case represent the summing point and takeoff point by a single node.

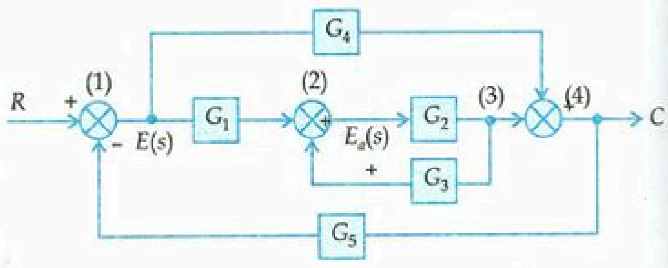
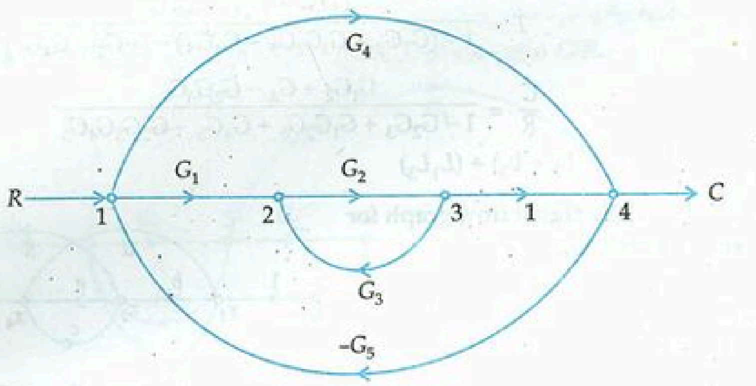


Fig. 1.104 (a)

3. If a summing point is placed after a takeoff point in the direction of signal flow, in such case,

represent the summing point and takeoff point by separate nodes connected by a branch having transmittance unity.

Consider the block diagram shown in Fig 1.104(*a*), the corresponding SFG is shown in Fig. 1.104 (*b*).



1.30. MASON'S GAIN FORMULA

The overall transmittance or graph transmittance between the source node and sink node is given by Mason's gain formula. Mason's gain formula for signal flow graph is as follows .

$$T = \frac{\Sigma g_k \Delta_k}{\Delta} \qquad ...(1.136)$$

where T = transfer function

Δ = 1 – [sum of all individual loop gain] + [sum of all possible gain products of two non-touching loops] – [sum of all possible gain products of three non-touching loops] + ...

g_k = gain of the k^{th} forward path

Δ_k = the part of Δ not touching the k^{th} forward path.

Δ_1 – the part of Δ not- touching forward path is 1 –

Consider the signal flow graph shown in Fig. (1.104(*b*)). There are two forward paths (i) 1-2-3-4 (*ii*) 1-4 therefore gain of the two paths will be

$$g_1 = G_1G_2$$
$$g_2 = G_4$$

There are three individual loops

$$L_1 = G_2G_3 \qquad (2\text{-}3\text{-}2)$$
$$L_2 = -G_1G_2G_5 \qquad (1\text{-}2\text{-}3\text{-}4\text{-}1)$$
$$L_3 = -G_4G_5 \qquad (1\text{-}4\text{-}1)$$

Since all three loops touching the forward path g_1, therefore $\Delta_1 = 1\text{-}0 = 1$ The first loop L_1 do not touch the forward path g_2, therefore $\Delta_2 = 1 - G_2G_3$

There are two non touching loops L_1 and L_3

$\Theta \qquad L_1L_3 = -G_2G_3G_4G_5$

Apply Mason's gain formula

$$T = \frac{C}{R} = \frac{g_1\Delta_1 + g_2\Delta_2}{\Delta}$$

$$T = \frac{G_1G_2 + G_4(1 - G_2G_3)}{1 - (G_2G_3 - G_1G_2G_5 - G_4G_5) - (-G_2G_3G_4G_5)}$$

$$\frac{C}{R} = \frac{G_1G_2 + G_4 - G_2G_3G_4}{1 - G_2G_3 + G_1G_2G_5 + G_4G_5 - G_2G_3G_4G_5}$$

Note : where $\Delta = 1 - (L_1 + L_2 + L_3) + (L_1L_3)$